Silent Data Error Investigator

A successful SQL query can still report the wrong revenue. This project trains a random forest to distinguish a real sales increase from double-counted orders and missing records.

Run in your browser

1. Open Train_in_Colab.ipynb via File > Upload notebook in Google Colab.

2. Connect to a CPU runtime. Choose Runtime > Run all.

3. When prompted, select Pipeline-Investigator-GitHub.zip. The notebook installs dependencies, trains a fresh model, prints four predictions and downloads the new results.

4. Open metrics.json in the downloaded results to inspect class-level scores and the confusion matrix.

How it works

The simulator creates orders, order items and payments in SQLite. An erroneous join repeats order totals for multi-item orders. A missing partition drops one order-ID bucket. Healthy examples represent ordinary demand and genuine growth.

Each simulated day produces four features: relative revenue change, relative order-count change, dashboard rows per unique order, and reported revenue divided by payments. A random forest with 80 trees and maximum depth 6 predicts the scenario from those four numbers.

Training and results

The training set has 800 examples. Evaluation uses 200 examples with distinct seeds and 50 per class. The submitted Colab run scored 200/200 with macro F1 of 1.00. Both splits use the same rules. Refunds, partial payments and unseen incidents require separate evaluation. The current implementation is a supervised ML baseline. An LLM investigation layer is future work.

Publish your project

1. Extract the download

Extract Pipeline-Investigator-GitHub.zip on your computer. Open the pipeline-investigator folder. Keep the ZIP for the Colab notebook.

2. Create the repository

Sign in to GitHub and select New repository. Name it pipeline-investigator. Choose Public for a portfolio or Private while developing. Leave the initial README option unchecked.

3. Upload the contents

Choose uploading an existing file on the empty repository page. Drag the contents of the extracted folder into the upload area, including saved_run. Commit the changes. README.md and the notebook belong at the repository root.

4. Check the result

Open the repository homepage. The formatted README should appear below the files. Open saved_run/metrics.json to see the submitted result. Open README.pdf to view this guide.

5. Reproduce the experiment

Follow the Colab instructions on page one. Retraining is optional because your original model and metrics are already included. A fresh run creates its own output archive.